

Internet Scan Source Identification: A Survey

Enhuan Dong^{*†}, Jiahai Yang^{*†}, Zhiliang Wang^{*†}, Xin He^{*}, Jie Fang^{*}, Shenao Li^{*}, Supei Zhang^{*}, Zeyuan Guo[§],
Hui Zhang^{*†}, Zimu Li[¶], Xiaoming Fu[‡]

^{*}Institute for Network Sciences and Cyberspace, Tsinghua University, Beijing, China

[†]Quan Cheng Laboratory, Jinan, China

[¶]IT Service Center, Tsinghua University, Beijing, China

[‡]Institute of Computer Science, University of Goettingen, Goettingen, Germany

[§]School of Computer Science, Beijing University of Posts and Telecommunications, Beijing, China

dongenhuan@tsinghua.edu.cn, yang@cernet.edu.cn, wzl@cernet.edu.cn, hexin21@mails.tsinghua.edu.cn,

fj22@mails.tsinghua.edu.cn, lsa22@mails.tsinghua.edu.cn, zsp23@mails.tsinghua.edu.cn,

1154459434@bupr.edu.cn, hzhang@cernet.edu.cn, lizimu@tsinghua.edu.cn, fu@cs.uni-goettingen.de

Abstract—Recently, Internet scanning activities have increased significantly, while the purposes and impacts of these scans have grown increasingly complex. Organizations dedicated to enhancing Internet security, as well as malicious ones, both conduct frequent and systematic scans across the entire IPv4 address space and partial IPv6 address space. In this context, the scan source identification, on which scan intelligence relies, is crucial as it supports in-depth analysis of the scanning behavior of organizations and helps uncover the real identity and size of potential attack organizations. In this article, we provide an extensive survey on Internet scan source identification. We summarize the common methodology and propose a taxonomy for classification. Based on this taxonomy, we outline the current state of Internet scan source identification research. Furthermore, we identify existing challenges and suggest future directions in this field.

I. INTRODUCTION

RECENTLY, scanning activities on the Internet have increased substantially. Network scanning refers to the process in which a host sends probing packets to gather various types of critical information within a network. The information includes details such as active hosts, open ports, software versions of services, operating system versions, *etc.* Each of these corresponds to different types of scanning, such as active host scanning, open port scanning, service scanning, and operating system scanning, *etc.* Common scanning tools include Nmap, ZMap, Masscan, ZGrab, *etc.* The leakage of the critical information may pose a threat to cyberspace security. For instance, scanning can reveal the version of a target's service software. When this information is cross-referenced with a public vulnerability database, it can expose software vulnerabilities, facilitating targeted network attacks. Furthermore, the malicious behavior of some well-known threats, such as the Conficker¹ and Mirai botnets², also contributes to Internet scanning activities.

About twenty years ago, research focused on detecting and preventing network scanning, as these activities often preceded attacks. Recently, however, the purpose and impact of scanning has evolved to become more complex. The security community

now increasingly recognizes the value of large-scale scans. Some emerging scan organizations openly share their objectives on public websites. We refer to them as *acknowledged* scan organizations³, *e.g.*, Censys⁴ and Shodan⁵. They conduct scans across the entire IPv4 address space with partial IPv6 address space, and some of their users may leverage the scan results to assess the security levels of their network assets or to conduct penetration testing, while others, *e.g.*, some researchers, may utilize them for large-scale analysis. Compared to the *unacknowledged* scan organizations, *i.e.*, those that do not share their objectives on public websites, the acknowledged ones are more likely to be non-malicious and contribute positively to the security community. Thanks to the advancements in efficient scanning techniques [2], including the emergence of scanning tools such as ZMap or Masscan, and the widespread availability of cloud server rental services, both acknowledged and unacknowledged scan organizations can perform frequent and large-scale scans of public address space, *e.g.*, Internet-wide scans.

Each scan organization employs multiple scanners. After thoroughly investigating existing research efforts, we define a scanner as an entity or program that is identified by distinct public IP addresses and send probing packets to one or more remote hosts with the objectives of (1) identifying hosts that respond to the packets and (2) gathering information from the response packets issued by the remote hosts. The specific definition of a scanner varies slightly depending on the focus of the relevant papers. Additionally, according to our definition, some complex attackers may incorporate scan behaviors into their broader activities, *e.g.*, they may attempt to log in once they find an SSH port (22) is open. In scanner-related research, these attackers are usually considered scanners and are not filtered out.

This article focuses on research efforts aimed at identifying the sources of the scanners, also referred to as scan sources. The sources of the scanners provide insights into the origins of their public IP addresses, which can be categorized at

¹<https://en.wikipedia.org/wiki/Conficker>

²[https://en.wikipedia.org/wiki/Mirai_\(malware\)](https://en.wikipedia.org/wiki/Mirai_(malware))

³We borrow the term “acknowledged” from [1].

⁴<https://search.censys.io>

⁵<https://www.shodan.io>

various levels, such as the organizational level, Internet Service Provider (ISP) level, country level, and so on.

The scan source identification enables targeted defenses against scanning. Recently, the scan organizations usually make their scanners behave like legitimate users to evade detection. Their scanners conduct porting scanning slowly by sending TCP SYN packets (*e.g.*, ZMap or Masscan in [2]), then continues to scan services (*e.g.*, ZGrab in [2]) after received TCP SYN-ACK packets. It is impossible to identify the scans based on packet header fields or payload content, except for a few acknowledged organizations scanning with well-known fingerprints at the IP header or in the payload. To defend such scan activities, related papers studies the methods to generate scan intelligence. Scan intelligence often includes both scanner IP addresses and their source information, facilitating targeted defenses against scanning from specific organizations, unacknowledged organizations, countries, ISPs, and others. With the aid of scan intelligence, network security devices such as firewalls and Network Intrusion Prevention Systems (NIPS) can enforce more fine-grained defenses, such as selectively dropping traffic from specific scan organizations.

Moreover, scan source identification allows for in-depth analysis of the scanning behavior of specific organizations. For instance, it can reveal the information each organization is collecting and the devices or IP addresses they are targeting, thus identifying potential attack targets (such as IP, port, service, or software). Establishing the relationships between scanning behaviors aids in analyzing new distributed attack patterns. Lastly, scan source identification helps uncover the real identity and size of potential attack organizations.

Related surveys can be categorized into two distinct groups: the first focuses on analyzing traffic from trap-based monitoring systems (*e.g.*, darknets) [3], while the second pertains to anomaly analysis based on network intrusion detection systems (NIDS) [4]. The first group comprises papers on scan source identification associated with the Conficker worm and several scanning botnets (*e.g.*, Sality SIP scan botnet) from earlier years; however, it excludes studies focused on identifying scan sources at the scan organization level and does not review the developed monitoring systems in recent years to capture scan traffic. The second group emphasizes the identification of scanning or reconnaissance behaviors, but does not address the determination of which scan sources the scanners originate from.

This survey addresses this gap in the literature by reviewing recent existing studies on scan source identification. We summarize the common methodology used in these studies and propose a taxonomy for categorizing them (§II). Based on this taxonomy, we describe the current state of scan source identification research (§§ III and IV). Furthermore, we identify open challenges and future directions in this field (§V).

II. METHODOLOGY AND TAXONOMY

After thoroughly investigating existing studies, we summarize the common methodology among them: they first deploy monitoring systems to obtain scan traffic, and then use various

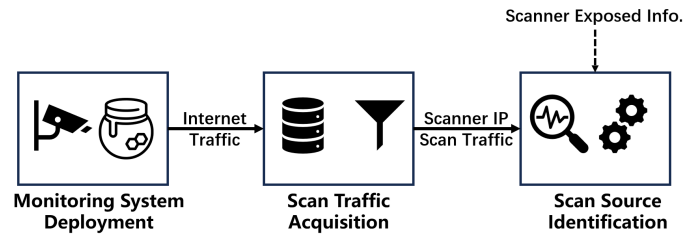


Fig. 1: Scan Source Identification Research Methodology

information exposed by scanners on the Internet to identify the sources to which scanners belong. Specifically, as presented in Fig. 1, researchers collect extensive Internet traffic, including scan traffic, through deployed monitoring systems. Then, they typically focus on a subset of the captured traffic, *i.e.*, the scan traffic, and designate the IP addresses of the traffic senders that reside in the external part of monitoring systems as scanner IPs. Finally, they identify the sources associated with these scanners using information exposed by the scanners on the Internet.

In this article, we provide a taxonomy of research works on scan source identification, depicted in Fig. 2. Scan source identification is at the root of the taxonomy (the first level in Fig. 2). We classify related research works according to the stages in the process of scan source identification (the second level in Fig. 2): the methods of scan traffic acquisition and the types of information used for identifying scan sources. We continue to categorize scan traffic acquisition methods based on the interactive capabilities of monitoring systems (the third level in Fig. 2). Additionally, we establish six branches to represent specific types of information used for identifying scan sources (the third level in Fig. 2). Finally, relevant papers are also listed in the third level.

III. SCAN TRAFFIC ACQUISITION

Monitoring systems commonly used to capture scan traffic can be divided into two categories: Non-interactive monitoring systems and interactive monitoring systems. The non-interactive monitoring systems do not interact with the scanners, while the interactive monitoring systems may send packets to the scanners as requested. Fig. 3 illustrates the interaction between different monitoring systems and scanners.

The typical non-interactive monitoring systems employed to capture scan traffic are Darknets, and Content Distribution Network (CDN) server firewalls.

Darknets [5], [6], [7], [8], [9], [1], [10]: Darknets passively collect traffic destined for IP addresses that are routable but unused. These addresses neither provide Internet services nor are used by actual Internet users. Any traffic destined to a darknet raises suspicion. Darknet traffic can be used to extract intelligence on scanning activities, DDoS attacks, misconfigurations, *etc.* It is noteworthy to mention that darknets do not reply any packets to the source IPs of received packets, which means darknet addresses cannot establish TCP connections because darknets do not respond with SYN-ACK to SYN packets. Additionally, the word darknet is known by several other terms, and “network telescopes” is widely used in earlier

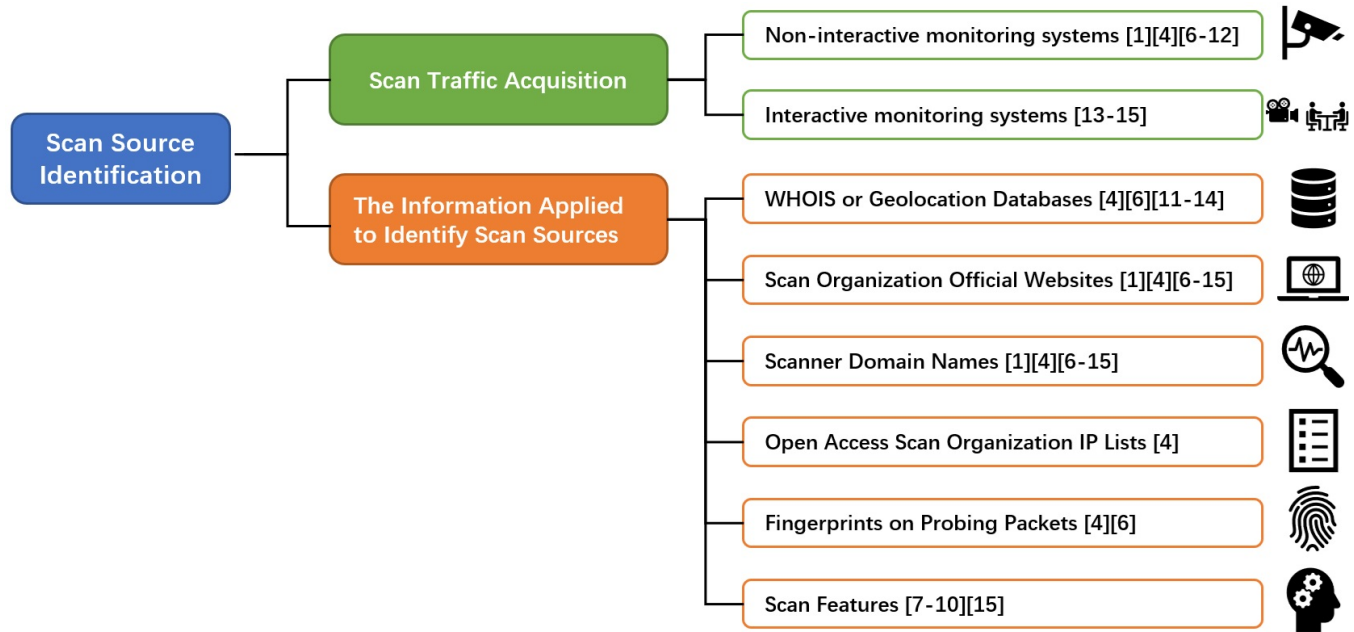


Fig. 2: Scan Source Identification Research Taxonomy

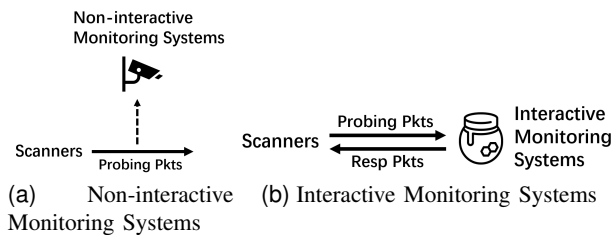


Fig. 3: The interaction between monitoring systems and scanners.

academic papers. However, in recent years, some studies [11], [12] have significantly enhanced the capabilities of network telescopes, which we will elaborate later.

CDN Server Firewalls [13], [14]: CDN server firewalls can also log scan traffic. The firewalls drop and log the packets that are not sent to any ports running active services. These packets are called “unsolicited” packets. Richter *et al.* [13], [14] regarded these firewalls as a distributed network telescope, and studied scanning activities according to these firewall logs. Different from darknets, the CDN servers offer real services, which enable them to log not only Internet-wide scans, *i.e.*, the scans destining random subsets of or the entire IPv4 address space, but also localized scans.

The interactive monitoring systems utilized to capture scan traffic are advanced network telescopes (*e.g.*, Spoki and DScope), and Honeypots.

Spoki [11]: Spoki is proposed as a reactive darknet. It responds with TCP SYN-ACK packets in real-time to TCP SYN packets received by the darknet. Therefore, Spoki can complete the TCP handshakes and capture more traffic sent by scanners to the darknet after the handshakes.

DScope [12]: DScope is designed as a cloud-based distributed network telescope. In cloud infrastructures, IP ad-

resses are assigned in bulk for brief periods, and due to the pseudorandom nature of these assignments, DScope can gather traffic on unoccupied IP addresses from a cloud provider. It accomplishes this by provisioning low-cost compute instances under those addresses and capturing incoming traffic. Like Spoki, DScope is also capable of accomplishing TCP handshakes.

Honeypots [15]: Similar to Darknet, Honeypot is a well-known traditional trap-based monitoring system. There are three primary categories of honeypots: low, medium, and highly interactive. A low-interactive honeypot is designed to engage with the initiator of the communication by mimicking basic protocols such as responding to Internet Control Message Protocol (ICMP) ECHO requests or replying to TCP SYN packets. For capturing scan traffic, low-interaction honeypots are adequate.

After capturing traffic from monitoring systems, depending on the problem of interest or the analysis to be performed, researchers may select traffic to obtain more “pure” scan traffic while minimizing the discard of traffic. The selection may directly identify scan traffic based on definitions provided by the researchers [5], [13], [14] or by filtering out non-scan traffic [6], [7], [9], [8].

At the end of this section, we would like to summarize recent advancements in network telescopes. Traditionally, network telescopes have been non-interactive, also known as darknets. Recently, researchers have enhanced these systems with additional capabilities, such as distribution [13], [14], [12], the ability to complete TCP handshakes [11], [12], cloud deployment [12], and the emulation of unresponsive application-layer services [11], [12]. These enhancements allow network telescopes to capture a broader range of scan traffic, offering a more comprehensive understanding of scanner behavior from multiple perspectives.

IV. THE INFORMATION APPLIED TO IDENTIFY SCAN SOURCES

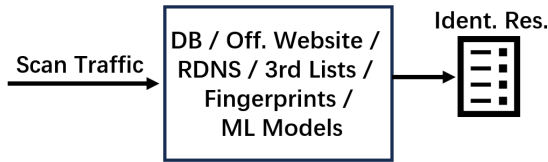


Fig. 4: The Workflow of Scan Source Identification Methods

The obtained scan traffic reveals the IP addresses and behaviors of the scanners. Research efforts have led to the development of various methods to identify the scan sources based on traffic data. The input of these methods is scan traffic, while the output is identification results. The workflow is summarized in Fig. 4.

A. IP Information from WHOIS or Geolocation Databases

After obtaining the scanner IP addresses, the most common method is to use IP information from WHOIS or geolocation databases to identify or describe their sources [5], [13], [14], [11], [12], [10]. This method can identify the scan source at a coarse granularity, such as ISP level, cloud computing company level, country level, *etc.* While such identification granularity may not disclose the specific organizations to which the scanner IPs belong, it is adequate to support preliminary analysis of the scan sources.

The WHOIS database is managed and maintained by Internet number assignment authorities, such as ICANN (Internet Corporation for Assigned Names and Numbers) and Regional Internet Registries (RIRs). These organizations are responsible for allocating and managing the registration information for IP addresses. The WHOIS database includes the Autonomous System (AS) numbers of the IP addresses, the organizations responsible for managing the IP addresses (such as ISPs, cloud computing companies, hosting providers, colocation centers, corporations, academic institutions, *etc.*), and the geographic locations of the IP addresses at the country or city level. Geolocation databases are specialized databases that map IP addresses to their geographical locations. These databases contain the most accurate possible information about the physical locations of devices connected to the Internet, often providing their longitude and latitude.

B. The Information from Scan Organization Official Websites

Three acknowledged scan organizations, Censys, Rapid⁶ and Driftnet⁷, make their scanner IP addresses public on their official websites. This information is frequently utilized by existing works [5], [13], [14], [11], [12], [6], [7], [8], [9], [15], [1], [10] to identify the scan organizations associated with scanner IP addresses. Unfortunately, to the best of our knowledge, only Censys, Rapid7 and Driftnet publicly share scanner IP addresses in this way.

⁶<https://opendata.rapid7.com>

⁷<https://internet-measurement.com/>

C. Scanner Domain Name Information

Some scan organizations register domain names for their scanner IP addresses. Reverse DNS (RDNS) resolving can reveal the domain names associated with scanner IP addresses. Further, these domain names may indicate their associated scan organizations. For example, some scanner IPs of the well-known cyberspace search engine, Shodan, have the same domain name pattern: *.shodan.io. RDNS resolving is also applied in many existing works [5], [13], [14], [11], [12], [6], [7], [8], [9], [15], [1], [10]. However, the number of scan organizations that have domain names for their scanner IP addresses is limited, and it is uncertain whether they have registered domain names for all of their scanner IP addresses. It is worth noting that RDNS resolving is also likely to identify the scan source at a coarser granularity. For instance, *.amazonaws.com is a possible RDNS resolved domain name pattern, indicating that the relevant IPs belong to AWS cloud computing company. However, it does not reveal the specific scan organizations these IPs belong to.

D. Open Access Scan Organization IP Information

Collins *et al.* [1] published open-access IP lists of scan organizations they had compiled. These lists draw on three sources. The first source, also the major one, consists of reports from third parties unconnected to either the authors or the scan organizations. These reports, submitted by external sources or gathered from blogs, articles, and emails, were verified by the authors to the best of their ability. The second source is information obtained directly from the official websites of scan organizations. Finally, the third source is based on RDNS resolution conducted by the authors themselves. Griffioen *et al.* [10] utilized these lists to analyze darknet traffic.

E. Fingerprint Information on Probing Packets

Some scan organizations may embed unique identifiers in their probing packets to signal their origin. For instance, HTTP probing packets from Censys contain payloads specific to Censys. Additionally, tools like Nmap, ZMap, Masscan, and Unicorn, as well as botnets like Mirai, exhibit well-known fingerprints [5], [10]. However, it is important to note that users can easily modify these fingerprints. Existing research indicates a decreasing number of scans that reliably match these fingerprints patterns [10]. Malicious organizations are more likely to modify the fingerprints of software such as ZMap, Masscan, and Mirai, as they often need to evade various identification techniques. In contrast, legitimate organizations tend to be more transparent about their scanning activities and do not attempt to conceal their intent.

F. Scan Feature Information

Given the limitations of the mentioned methods of identifying scan sources, some studies [6], [7], [8], [9], [15] are dedicated to identifying the scan organizations to which the scanners belong by extracting features from the scan traffic. With the help of these scan features and machine learning (ML) techniques, they can identify the scan organizations with

which more scanners are associated and analyze the IP clusters of unknown scan organizations. The main contribution of these studies is to identify scan organizations with scan feature information; however, it is worth noting that their methods also rely on the information mentioned in the previous subsections to label the ground-truth scan organizations for some of the scanner IP addresses. The methods presented in the previous subsections are rule-based methods. They rely on predefined signatures or expert knowledge, making them easier to evade, whereas ML-based methods can learn patterns from data, generalize to unseen behaviors, and adapt dynamically with incremental learning.

TABLE I: Scan Feature Comparison

Literature	Scan Features
DANTE [6]	The sequences of ports contacted by scanners
DarkVec [7] and i-DarkVec [9]	The sequences of the scanner IP addresses that reach each port set
Kallitsis's Framework [8]	The total number of packets sent, total bytes, average inter-arrival time between sent packets, number of distinct targeted ports ...
Scanner-Hunter [15]	The sequence number of each Modbus request packet within a TCP connection, whether the Modbus request packet has been responded to ...

DANTE [6]: DANTE is aimed at analyzing darknet traffic and it regards the sequences of ports contacted by scanners as sequences of words in a Natural Language Processing (NLP) problem. First, DANTE depicts each scanner by the sequence of ports it contacts within a time interval. Then, for each port, DANTE trains a separate Word2Vec embedding. By averaging the embeddings of the ports that the scanner has targeted, each scanner IP is represented as a feature vector. Cohen et al. [6] believed that these feature vectors could capture the scanners' intent. Finally, since the scanners of the same scan organization may have similar intent, a typical clustering algorithm, DBSCAN, is used to cluster the feature vectors of the scanner IP addresses.

DarkVec [7]: DarkVec is also designed for analyzing darknet traffic. It utilizes NLP techniques and begins by forming a corpus. It builds multiple target port sets, and treats the temporal sequences of the IP addresses of the scanners that reach each port set within a time interval as sequences of words. Then, DarkVec trains a single Word2Vec embedding that maps each scanner IP to a feature vector. Finally, k-NN based semi-supervised and unsupervised ML models are employed to identify the scanners associated with the same scan organization. DarkVec has significantly faster training times compared to DANTE.

i-DarkVec [9]: i-DarkVec is an incremental version of DarkVec, and it differs from DarkVec solely in the training of Word2Vec models. In reality, scan traffic does not arrive at the darknet all at once. Instead, it increases gradually over time. DarkVec is not an incremental solution, which means that every update necessitates retraining the entire corpus. For i-DarkVec, at each timestep t_{i+1} , a new batch of scan traffic arrives, from which the corpus is generated, and the embeddings are incrementally updated using the weights computed at the t_i timestep. i-DarkVec offers two main advantages: (i) it significantly reduces the corpus size, thereby speeding

up the Word2Vec learning, and (ii) it allows the training to automatically prioritize newer information.

Kallitsis's Framework [8]: Kallitsis's Framework is also proposed for darknet traffic. First, it extracts features for each scanner based on the daily scan traffic. The scan features include the total number of packets sent, total bytes, average inter-arrival time between sent packets, number of distinct targeted ports, number of distinct targeted addresses, the ratio of the number of scanners within a routing prefix over all IP addresses covered by that prefix, whether the destination IPs and IPID constant, random or fixed increasing, whether any TCP options set in TCP-related scan, set of targeted ports, protocol request types (*e.g.*, TCP SYN, UDP, *etc.*), and the set of TTL values. Then, Kallitsis's Framework uses deep autoencoders to project the high-dimensional feature vectors into low-dimensional feature vectors. Finally, K-means, a typical clustering algorithm, is used to cluster the feature vectors of the scanners.

Scanner-Hunter [15]: Scanner-Hunter is designed for Modbus honeypot traffic. The Modbus protocol is a TCP-based industrial control protocol. Some scan organizations conduct Modbus service scans to discover Internet-accessible ICS devices. The Modbus honeypots in [15] support multiple Modbus protocol interactions with scanners in one TCP connection. Scanner-Hunter extracts features for each traffic session, which is defined as several consecutive TCP connections with the same source IP, destination IP and transport protocol within a time interval. Scanner-Hunter employs a set of feature vectors to represent each traffic session. An individual feature vector describes each Modbus request packet. Since a single traffic session may contain multiple Modbus request packets distributed across several TCP connections, the feature vectors describing these request packets constitute that traffic session's set of feature vectors. Each feature vector is composed of the sequence number of each Modbus request packet within a TCP connection, whether the Modbus request packet has been responded to, the length of a Modbus request packet, the total number of Modbus request packets within a TCP connection, the serial number of a TCP connection within the traffic session, the total number of TCP connections within the traffic session, the fields "Transaction Identifier", "Unit Identifier" and "Function Code" of the Modbus protocol, and the "Unit Identifier" of the previous packet. Then, Scanner-Hunter uses a novel self-expanding multi-class classification model to identify scan organizations. This model consists of a One-vs-One model, a Random Forest model, a self-updating mechanism, a self-expanding mechanism, IP prefix judgment, *etc.* Besides Modbus honeypot traffic, Scanner-Hunter also needs open-source Modbus scan tools to generate traffic as the input of the whole identification system.

V. CHALLENGES AND FUTURE DIRECTIONS

This section discusses both Internet-wide and localized scans, outlines the open challenges faced by existing research works and identifies future directions for each challenge.

A. Internet-wide Scans

Effectively capturing scan traffic is challenging: As introduced in §III, non-interactive and interactive monitoring systems are applied to collect scan traffic. However, they both have drawbacks. The information collected by non-interactive monitoring systems is limited. As these monitoring systems do not establish TCP connections with the scanners, service scan traffic is hard to be captured. Service scan traffic, however, is valuable for scan source identification. Since services can be hosted on unexpected ports, scanners may initiate service scans for ports beyond those assigned by IANA, therefore, the service scans can be quite complex and diverse. The scanners belonging to the same organization may have similar features in service scan behavior. Since non-interactive monitoring systems, particularly darknets and CDN server firewalls, cannot capture service scan traffic, related research cannot use service scan behavior as a basis for analyzing the organizations of scanners. As for interactive monitoring systems, they are easier for scanners to identify and eventually bypass, making scanners avoid sending packets to the monitoring systems, *e.g.*, Shodan can identify honeypots.

To address the shortcomings of both non-interactive and interactive monitoring systems, it is worthwhile to study covert interactive monitoring systems, which are designed to be difficult to identify and capable of capturing service scan traffic. Though some early studies on anti-identification honeypots have emerged, the proposed methods remain insufficiently mature, often resulting in inefficient data collection, and have yet to be widely deployed.

Scan source identification at organization level is challenging: As illustrated in §IV, identifying scan sources at a coarse granularity, such as the ISP or country level, is not difficult. However, identifying scan sources at the organizational level is much more meaningful but also significantly more challenging. Though some solutions have been proposed as shown in §IV, they still face the following challenges.

The challenges in scan source identification can be grouped into three major categories. Firstly, the dynamic nature of scan organizations poses significant difficulties. These organizations frequently change their scanner IPs by rotating among different leased cloud hosts, making it difficult to track their activities over time. In addition to this IP-hopping strategy, they employ various evasion techniques to further obscure their identity. For instance, they may mimic the behavioral characteristics of other scan organizations, making it challenging to distinguish them based on existing ML models. Furthermore, as discussed in §IV-E, they can modify the fingerprints of their probing packets—such as header fields, or payload structures—to evade identification. Additionally, the growing number of scan organizations further exacerbates this issue. Secondly, current identification techniques are constrained by several limitations. While existing studies [6], [7], [8], [9], [15] can uncover new clusters of scanner IPs, they fail to associate these clusters with specific scan organizations without ground truth information. Furthermore, these methods heavily depend on manual identification, which is inefficient and lacks automation. Lastly, challenges related to feature design and robustness

further hinder progress. The robustness of features, such as timing properties of scanning ports and statistical attributes of scanning activities, has not been thoroughly evaluated [6], [7], [8], [9]. Moreover, these features are influenced by specific data sources and monitoring systems, introducing variability and limiting their general applicability.

Future solutions should be aligned with the three major categories of issues and fully leverage emerging techniques, particularly ML, to enhance the accuracy and efficiency of scan source identification. First, to address the dynamic nature of scan organizations, ML models must be continuously updated to keep pace with these dynamics. Therefore, it is essential to adopt carefully selected ML models that support incremental updates, enhance recognition accuracy, and minimize the training cost for each identification cycle. Second, to address the limitations of current identification techniques, in addition to exploring innovative approaches to obtain reliable ground truth for associating scanner IP clusters with specific organizations, future research should focus on implementing automated open-set recognition models. These models can enhance automation by distinguishing between known and unknown scanning behaviors without manual intervention. Finally, to tackle challenges related to feature design and robustness, ML techniques should be employed to automate the extraction of robust features, ensuring adaptability across diverse monitoring environments. Additionally, more advanced ML models should be leveraged to improve identification performance even when relying on less robust features, thereby enhancing the overall reliability of the identification process.

B. Localized Scans

Scan traffic identification and scan source identification are more challenging for localized scans: Localized scans are very common [13], [11]. For Internet-wide scans, effective defense can be achieved if the monitoring systems receive the probing packets before they reach the protected network, scan intelligence is generated promptly, and the intelligence is sent to the network defense devices of the protected network. Effective defense against Internet-wide scans relies on the timely generation of intelligence.

However, as for localized scans, if the adversary uses a new scanner IP each time it scans the target network (*i.e.*, increasing the cost of the attack), as far as we know, there is no ideal way to defend against this type of scan. For critical IPs, whitelist techniques can be used to protect critical information from localized scans, but this method may impact the normal use of services associated with the IP and is not feasible for open services.

Future solutions should prioritize the defense against localized scans, with particular emphasis on protecting the critical information of open services. Low-cost protection measures must also be considered. We believe that deploying monitoring systems within protected networks will enhance defense against localized scans, as intelligence obtained from external sources may not be applicable to one's own network. Additionally, the challenges and future directions discussed in the previous subsection are equally relevant to localized scans.

VI. CONCLUSION

As large-scale Internet-wide scans become more common, identifying scan sources is increasingly crucial. This survey is the first to focus on this topic. We review the methodologies used in related studies and propose a taxonomy for classification. Based on this taxonomy, we discuss the current research on scan source identification. Additionally, we highlight existing challenges and suggest future research directions for both Internet-wide and localized scans.

ACKNOWLEDGMENT

This work is supported by the National Key R&D Program of China under Grant 2022YFB3102902, and the National Natural Science Foundation of China under Grant 62172251. Jiahai Yang is the corresponding author of this paper.

REFERENCES

- [1] M. Collins, A. Hussain, and S. Schwab, "Identifying and Differentiating Acknowledged Scanners in Network Traffic," in *IEEE EuroS&PW*, 2023, pp. 567–574.
- [2] Z. Durumeric, D. Adrian, A. Mirian, M. Bailey, and J. A. Halderman, "A Search Engine backed by Internet-wide Scanning," in *ACM CCS*, 2015, pp. 542–553.
- [3] J. Saleem, R. Islam, and Z. Islam, "Darknet Traffic Analysis: A Systematic Literature Review," *IEEE Access*, 2024.
- [4] K. He, D. D. Kim, and M. R. Asghar, "Adversarial Machine Learning for Network Intrusion Detection Systems: A Comprehensive Survey," *IEEE Communications Surveys & Tutorials*, vol. 25, no. 1, pp. 538–566, 2023.
- [5] Z. Durumeric, M. Bailey, and J. A. Halderman, "An Internet-Wide View of Internet-Wide Scanning," in *USENIX Security*, 2014, pp. 65–78.
- [6] D. Cohen, Y. Mirsky, M. Kamp, T. Martin, Y. Elovici, R. Puzis, and A. Shabtai, "DANTE: A framework for mining and monitoring darknet traffic," in *Computer Security—ESORICS, Proceedings, Part I* 25. Springer, 2020, pp. 88–109.
- [7] L. Gioacchini, L. Vassio, M. Mellia, I. Drago, Z. B. Houidi, and D. Rossi, "DarkVec: Automatic Analysis of Darknet Traffic with Word Embeddings," in *ACM CoNEXT*, 2021, pp. 76–89.
- [8] M. Kallitsis, R. Prajapati, V. Honavar, D. Wu, and J. Yen, "Detecting and Interpreting Changes in Scanning Behavior in Large Network Telescopes," *IEEE TIFS*, vol. 17, pp. 3611–3625, 2022.
- [9] L. Gioacchini, L. Vassio, M. Mellia, I. Drago, Z. B. Houidi, and D. Rossi, "i-DarkVec: Incremental Embeddings for Darknet Traffic Analysis," *ACM TOIT*, vol. 23, no. 3, pp. 1–28, 2023.
- [10] H. Griffioen, G. Koursiounis, G. Smaragdakis, and C. Doerr, "Have you SYN me? Characterizing Ten Years of Internet Scanning," in *ACM IMC*, 2024.
- [11] R. Hiesgen, M. Nawrocki, A. King, A. Dainotti, T. C. Schmidt, and M. Wählisch, "Spoki: Unveiling a New Wave of Scanners through a Reactive Network Telescope," in *USENIX Security*, 2022, pp. 431–448.
- [12] E. Pauley, P. Barford, and P. McDaniel, "DScope: A Cloud-Native Internet Telescope," in *USENIX Security*, 2023, pp. 5989–6006.
- [13] P. Richter and A. Berger, "Scanning the Scanners: Sensing the Internet from a Massively Distributed Network Telescope," in *ACM IMC*, 2019, pp. 144–157.
- [14] P. Richter, O. Gasser, and A. Berger, "Illuminating Large-scale IPv6 Scanning in the Internet," in *ACM IMC*, 2022, pp. 410–418.
- [15] C. Sheng, Y. Yao, L. Zhao, P. Zeng, and J. Zhao, "Scanner-Hunter: An Effective ICS Scanning Group Identification System," *IEEE TIFS*, 2024.

Enhuan Dong received the B.E. (2013) degree from Harbin Institute of Technology, Harbin, China, and the Ph.D. (2019) degree from Tsinghua University, Beijing, China. He was a visiting Ph.D. student at the University of Goettingen in 2016–2017. He is currently an Assistant Research Professor in the Institute for Network Sciences and Cyberspace, Tsinghua University. His research interests include network security, network operations and network transport.

Jiahai Yang received the B.Sc. degree in computer science from Beijing Technology and Business University, and the M.Sc. and Ph.D. degrees in computer science from Tsinghua University, Beijing, China. He is currently a Professor with Institute for Network Sciences and Cyberspace, Tsinghua University. His research interests include network management, network measurement, network security, Internet routing, cloud computing, and big data applications.

Zhiliang Wang received the B.E., M.E. and Ph.D. degrees in computer science from Tsinghua University, China in 2001, 2003 and 2006 respectively. Currently he is an Associate Professor in the Institute for Network Sciences and Cyberspace at Tsinghua University. His research interests include formal methods and protocol testing, next generation Internet, network measurement.

Xin He is currently a PhD candidate at Institute for Network Sciences and Cyberspace, Tsinghua University. His research interests include network security and network management.

Jie Fang is currently a postgraduate student of Institute for Network Sciences and Cyberspace, Tsinghua University. His research interests include network measurement and network security.

Shenao Li is now working on his master's degree at Institute for Network Sciences and Cyberspace, Tsinghua University. His research interests include network measurement, network security, and Internet routing.

Supeí Zhang is currently pursuing his master's degree at Institute for Network Sciences and Cyberspace, Tsinghua University. His research interest is network security.

Zeyuan Guo is currently pursuing a Bachelor's degree in Computer Science and Technology at Beijing University of Posts and Telecommunications. His research interests include network security and wireless communications.

Hui Zhang is currently a Senior Engineer with Institute for Network Sciences and Cyberspace, Tsinghua University. His research interests include network management, network measurement, network security, and Internet routing.

Zimu Li is currently a Senior Engineer with IT Service Center of Tsinghua University. His research interests include IoT management and network security.

Xiaoming Fu received his Ph.D. in computer science from Tsinghua University, Beijing, China in 2000. He has been a professor in computer science and heading the Computer Networks Group since 2007. Prof. Fu's research interests include network architectures, protocols, and applications. He is an IEEE Fellow, an ACM Distinguished Member, a fellow of IET and member of the Academia Europaea.